

Supplementary Information: Ephaptic A β -to-C-Fiber Crosstalk and Multi-Fiber Spatial Summation

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S1 Numerical verification

n	lagged, $\Delta t = 2.5 \mu s$	lagged, $\Delta t = 0.125 \mu s$	monolithic, $\Delta t = 5 \mu s$	monolithic, $\Delta t = 0.125 \mu s$
1	1.39	1.51	1.54	1.51
10	5.01	10.00	10.03	9.94
25	6.10	15.41	15.39	15.47
50	6.28	20.88	21.22	21.72

Table S1. The lagged coupling scheme vs the monolithic scheme, in the extended-compartment configuration (HH $E_{Na} = +35.64$ mV, 100-nA stimulus inside the compartment, $\Delta z = 10 \mu m$). Entries are the peak C-fiber depolarization between $z = 3$ and 8 mm (mV). The lagged error grows with n ; the monolithic values are Δt -independent to within 2.4 %.

model	n	$\Delta z 20 \mu m$	$\Delta z 10 \mu m$	production (5 μm , 1 μs)	$\Delta z 1.25 \mu m$	$\Delta t 0.125 \mu s$	conductance-implicit
HH	1	1.43	1.53	1.56	1.57	1.55	1.54
HH	10	10.25	10.79	10.93	10.97	10.93	10.89
HH	25	18.88	19.12	19.34	19.40	19.39	19.41
HH	50	22.35	23.26	23.51	23.59	23.62	23.76
NavC	1	1.43	1.53	1.56	1.57	1.55	1.54
NavC	10	10.25	10.79	10.93	10.97	10.93	10.89
NavC	25	18.90	19.13	19.35	19.41	19.40	19.42
NavC	50	22.36	23.28	23.53	23.61	23.63	23.78

Table S2. Grid and time-step refinement of the peak C-fiber depolarization in the lesion (mV) in the production configuration. Unless stated, $\Delta z = 5 \mu m$, $\Delta t = 1 \mu s$, explicit ionic currents. “Conductance-implicit”: ionic conductances in the system matrix.

S2 Membrane verification

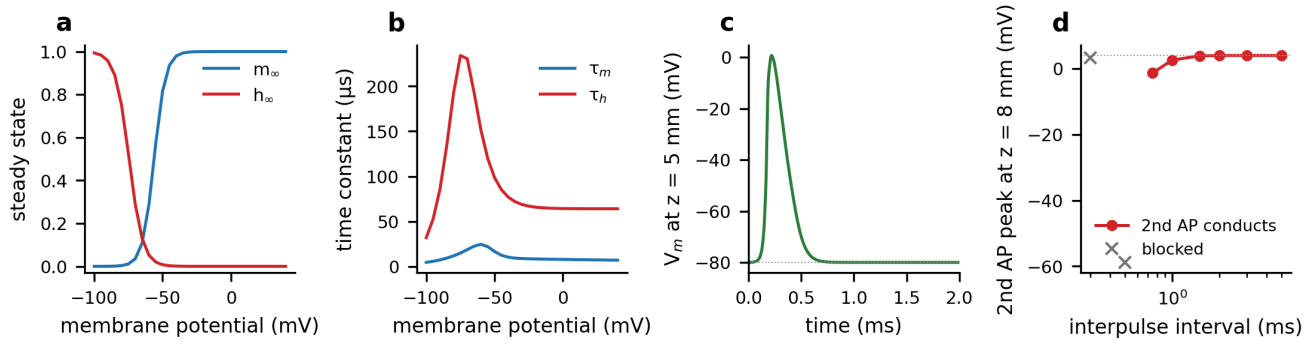


Fig. S1 Verification of the A β source. (a) CRRSS steady-state activation and inactivation and (b) their time constants over the modelled range, at 37 °C. (c) Action potential at a node ($z = 5$ mm) followed through repolarization to rest. (d) Paired-pulse recovery: peak of the response at $z = 8$ mm after a second stimulus, against the interpulse interval. Grey crosses mark intervals at which no second action potential propagates; the value plotted there is the decaying tail of the first action potential, not a second spike.

interval (ms)	APs at $z = 8$ mm	2nd AP peak (mV)	2nd AP CV (m/s)	2nd AP conducts
0.30	1	+3.4	–	False
0.50	1	-58.9	–	False
0.75	2	-1.3	36.0	True
1.00	2	+2.6	39.2	True
1.50	2	+3.9	40.4	True
2.00	2	+4.0	40.4	True
3.00	2	+4.0	40.0	True
5.00	2	+4.0	40.0	True

Table S3. Paired-pulse recovery of the isolated A β fiber ($2 \times$ threshold, 0.2-ms pulses at node 0; counts and peaks read at $z = 8$ mm). The first action potential conducts at 40.0 m/s in every run.

C-fiber membrane	rest (mV)	τ_m at rest (ms)	λ , small signal (μm)	R_{in} predicted ($\text{M}\Omega$)	R_{in} measured ($\text{M}\Omega$)	rheobase, 1 ms (nA)
HH	-65.00	1.48	198	68.9	68.8	0.16
NavC	-66.82	1.37	176	61.5	61.3	0.70

Table S4. Passive properties of the isolated C-fiber membranes under the production geometry. λ and R_{in} are small-signal values, computed from the slope conductance dI_{ion}/dV at rest; R_{in} measured is the steady response to a 1-pA injection at $z = 5$ mm. Rheobase is for a 1-ms point injection at the same site.

S3 Strength–duration characterization

C-fiber membrane	Q _{th} 0.1 ms (pC)	V _{peak} 0.1 ms (mV)	V _{peak} 1 ms (mV)	V _{peak} 20 ms (mV)	I _{th} 0.1 / I _{th} 20 ms
HH, 6.3 °C	0.14	-30	-51	-53	21.3
HH, 25 °C	0.10	-38	-43	-38	5.4
HH, conductances × 5	0.11	-38	-54	-56	12.7
HH, rates & conductances × 5	0.07	-48	-52	-53	4.6
HH, rates & conductances × 10	0.05	-51	-53	-53	2.5
NavC, $\tau_{m8} = 1.5$ ms	0.41	39	-19	-31	7.3
NavC, $\tau_{m8} = 0.05$ ms	0.16	-24	-34	-39	4.0
NavC, time constants ÷ 10, conductances × 10	0.22	-19	-31	-31	1.2

Table S5. Point current injection at $z = 5$ mm into the isolated C-fiber. Q_{th} : threshold charge for a 0.1-ms pulse; V_{peak} : largest membrane potential at the injection site for a pulse 0.5 % below threshold; last column: ratio of threshold currents for 0.1-ms and 20-ms pulses.

S4 Sensitizing bias

model	bias (A/m ²)	rest (mV)	spontaneous AP	peak V, n = 25 (mV)	peak V, n = 100 (mV)	evoked AP (any n)
HH	0	-65.00	no	-45.7	-38.1	no
HH	0.01	-64.20	no	-44.9	-37.3	no
HH	0.02	-63.48	no	-44.2	-36.6	no
HH	0.03	-62.85	no	-43.5	-35.9	no
HH	0.04	-62.27	no	-43.0	-35.4	no
HH	0.05	-61.73	no	-42.4	-34.8	no
HH	0.06	-61.24	no	-41.9	-34.4	no
HH	0.07	-60.78	no	-41.5	-33.9	no
NavC	0	-66.82	no	-47.5	-39.9	no
NavC	0.25	-59.30	no	-40.0	-32.3	no
NavC	0.5	-56.11	no	-36.9	-29.1	no
NavC	1	-52.20	no	-33.1	-25.2	no
NavC	1.5	-49.52	no	-30.5	-22.6	no
NavC	2	-47.39	no	-28.4	-20.6	no
NavC	2.5	-45.55	no	-26.7	-18.8	no
NavC	3	-43.90	no	-25.1	-17.3	no
NavC	3.5	-42.35	no	-23.7	-15.9	no

Table S6. Uniform depolarizing bias current on the C-fiber (production configuration). “Spontaneous AP”: no-stimulus control of 200 ms. Peak V: largest C-fiber membrane potential in the lesion after the A β volley. The HH resting state is linearly stable up to 0.095 A m⁻² and unstable from 0.1 A m⁻²; the NavC resting state stays stable over the whole tested range (to 4 A m⁻²) and a second stable, depolarized state appears above 2.75 A m⁻² (space-clamped analysis with linear stability from the Jacobian, e10_equilibria.csv).

S5 Temporal dispersion: convergence in the number of onset phases

study	K	Δz (μm)	spacing $W/(K-1)$ (μs)	peak ΔV (mV)
phases K	11	20	150	2.23
phases K	21	20	75	1.31
phases K	41	20	38	0.97
phases K	61	20	25	0.91
phases K	81	20	19	0.89
grid Δz	41	20	38	0.97
grid Δz	41	10	38	1.02
grid Δz	41	5	38	1.03

Table S7. Dispersed volley ($n = 25$, $W = 1.5$ ms, NavC). The peak depends on the number of onset phases K until the spacing $W/(K - 1)$ is well below the ~ 0.1 -ms duration of one fiber's contribution; the production runs use a spacing of at most $25 \mu\text{s}$. Grid refinement at $K = 41$ changes the peak by 6.2 %, so the sweep is run at $\Delta z = 20 \mu\text{m}$. One classification does depend on that choice: the synchronous $n = 25$ volley is close to A β conduction block, and on $\Delta z = 20 \mu\text{m}$ its action potentials still cross the lesion, slowly (13 m/s), where every grid from $\Delta z = 10 \mu\text{m}$ down blocks them (Table S2). The peak depolarization differs by only 2.3 % between the two grids, and the attenuations quoted in the text are ratios taken within one grid, so neither the attenuation nor the primary endpoint is affected.

S6 The extended-compartment configuration

electrode current	model	A β stimulus (nA)	propagating C-fiber AP for $n \geq$	initiation (mm)	A β node 0 peak, $n = 25$ (mV)	max
omitted	HH	3.02	none ($n \leq 50$)	–	+17	71
omitted	HH	10	25	0.00	+110	102
omitted	HH	30	10	0.00	+413	276
omitted	HH	100	2	0.00	+1465	935
omitted	NavC	3.02	none ($n \leq 50$)	–	+17	71
omitted	NavC	10	none ($n \leq 50$)	–	+110	102
omitted	NavC	30	none ($n \leq 50$)	–	+414	277
omitted	NavC	100	none ($n \leq 50$)	–	+1466	938
returned	HH	3.02	none ($n \leq 50$)	–	+4	34
returned	HH	10	none ($n \leq 50$)	–	+24	65
returned	HH	30	none ($n \leq 50$)	–	+139	194
returned	HH	100	25	0.07	+556	676
returned	NavC	3.02	none ($n \leq 50$)	–	+4	34
returned	NavC	10	none ($n \leq 50$)	–	+24	65
returned	NavC	30	none ($n \leq 50$)	–	+139	194
returned	NavC	100	50	0.07	+556	676

Table S8. Shared compartment along the whole cable, sealed ends, A β electrode at node 0 inside the compartment ($\kappa = 10^9 \text{ m}^{-2}$, $A_e = 16.4 \mu\text{m}^2$). C-fiber action potentials are launched at the electrode site only with suprathreshold electrode currents; the electrode-site membrane potentials show the non-physiological state produced by 100 nA.

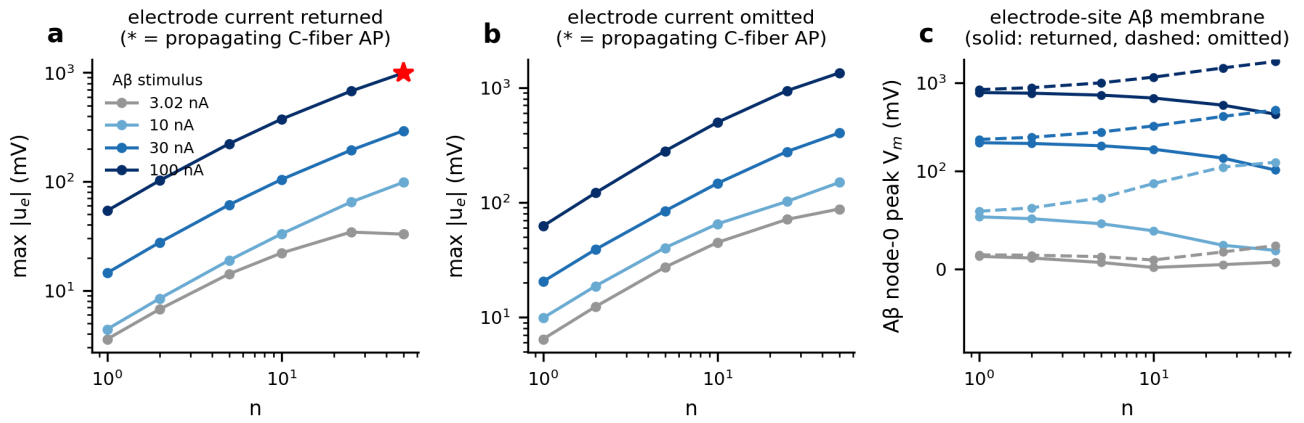


Fig. S2 Earlier full-length geometry. (a, b) Peak C-fiber membrane potential at the electrode site $z = 0$ vs n for A β stimuli of 2 $\times$ threshold to 100 nA (stars: propagating C-fiber AP launched at the electrode). (c) Peak A β membrane potential at the stimulated node.

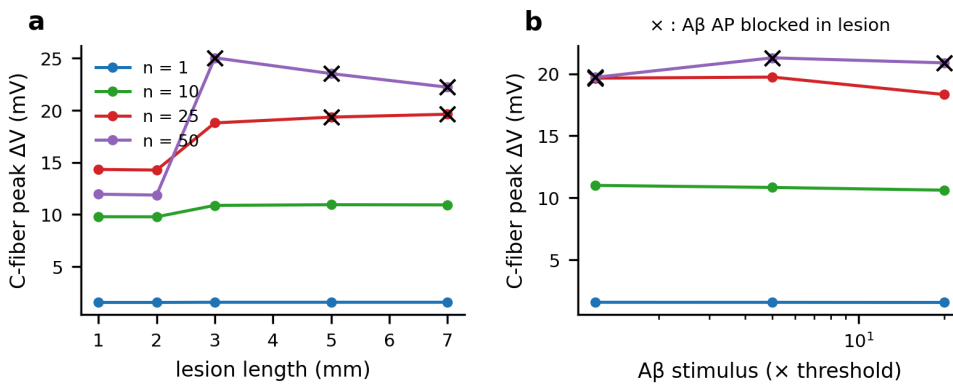


Fig. S3 Lesion length (a) and A β stimulus strength (b) in the production configuration (NavC). No C-fiber action potential occurred.